

THERMAL ENGINEERING -I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2×10=20

- (i) First law of thermodynamics says about
 (a) Quality of energy (b) quantity of energy (c) both of (a) & (b) (d) none of these
- (ii) Which of the following is the first law for a closed system undergoing a cycle?
 (a) $\oint dW = \oint dQ$ (b) $\oint dW = -\oint dQ$ (c) $\oint dW = \oint dQ$ (d) None of the mentioned
- (iii) A heat engine produces an output of 12 kW by rejecting 23 kJ/sec of heat to a sink at a particular temperature. The efficiency of the engine will be (a) 33.4 % (b) 65.7% (c) 34.3% (d) 52.17%
- (iv) In a throttling device, what do we get as SFEE when changes in PE and KE are taken zero?
 (a) $dQ/dm \neq 0$ (b) $dW/dm \neq 0$ (c) $h_1 = h_2$ (d) none of the mentioned
- (v) Coefficient of performance(COP) is defined as
 (a) heat leakage/work input (b) work input/heat leakage
 (c) refrigeration effect/work input (d) work input/latent heat of condensation
- (vi) What is the dryness fraction (x) for saturated water, when water just starts boiling?
 (a) x = 0 (b) x = 1 (c) x = 0.9 (d) x = 0.5
- (vii) Entropy is a
 (a) path function, intensive property (b) path function, extensive property
 (c) point function, intensive property (d) point function, extensive property
- (viii) Triple point of a pure substance on P-V diagram is represented by a
 (a) point (b) line (c) curve (d) triangle
- (ix) At a pressure below the triple point line,
 (a) the substance cannot exist in the liquid phase
 (b) the substance when heated transforms from solid to vapour
 (c) both of the mentioned (d) none of the mentioned
- (x) The amount of heat transferred to convert unit mass of solid to vapour or vice versa is called as
 (a) latent heat of vapourization (b) latent heat of fusion
 (c) latent heat of sublimation (d) specific heat
- (xi) The function of a nozzle is to?
 (a) decrease the velocity of a fluid at the cost of its pressure gain
 (b) increase the velocity of a fluid at the cost of its pressure drop
 (c) increase the velocity of a fluid and also its pressure
 (d) none of the mentioned.
- (xii) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is (a) - 90 kJ (b) -30 kJ (c) 30 kJ (d) 90 kJ
- (xiii) Mollier diagram is a plot of
 (a) temperature and entropy (b) enthalpy and entropy
 (c) pressure and enthalpy (d) pressure and volume
- (xiv) What is the function of economizer?

- (a) It is used to heat the feed water (b) It is used to improve the quality of steam
 (c) It is used to heat air being supplied for combustion
 (d) It separates suspended water particles from steam
- (xv) A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m^3 to 0.4 MPa, 0.03 m^3 in a process following $PV^n = \text{const.}$ The index of compression is
 (a) 1.23 (b) 1.337 (c) 1.4 (d) 1.46

GROUP-B

Answer any Five (05) questions.

2. a) Define the following terms related to a thermodynamic system: Path of a process, Extensive property and & Point function. ①
 What do you understand by property of a system? $[(1 \times 3) + 1]$
- b) Prove that for a reversible adiabatic process, $PV^\gamma = \text{constant}$. ① (4)
3. a) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW? (6) ②
- b) State Clausius inequality. (2)
4. a) State Charles' law. (1)
- b) What is PMM-I? Is it possible to devise a PMM-I? Justify your answer. $(1+1+1)$
- c) A house requires $2 \times 10^5 \text{ kJ/h}$ for heating in winter. Heat pump is used to absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is $3 \times 10^4 \text{ kJ/h}$. Determine (i) Heat abstracted from outside and (ii) Co-efficient of performance. $(2+2)$
5. a) Derive the relationship between temperature and volume for an ideal gas, if it undergoes a polytropic process. <https://www.pyqonline.com> (2)
- b) A gas undergoes a thermodynamic cycle consisting of the following processes:
 - Process 1-2: Constant pressure $p = 1.4 \text{ bar}$, $V_1 = 0.028 \text{ m}^3$, $W_{12} = 10.5 \text{ kJ}$
 - Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$
 - Process 3-1: Constant volume, $U_1 - U_3 = -26.4 \text{ kJ}$. There are no significant changes in KE and PE.
 (i) Sketch the cycle on a p-V diagram (ii) Calculate the net work for the cycle in kJ. ①
 (iii) Calculate the heat transfer for process 1-2 (iv) Show that $\Sigma Q = \Sigma W$. $(1+2+1+2)$
6. a) Draw the phase equilibrium diagram on p-v & T-s coordinates considering water as a pure substance. Indicate all relevant constant property lines in the plot. (2+2)
- b) Find the specific enthalpy and specific entropy of steam when the pressure is 2 MPa and the specific volume is $0.09 \text{ m}^3/\text{kg}$. (2+2)
7. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in m of water column. (all the symbols have their usual meanings). (5)
- b) Draw a neat sketch of a horizontal shell, water tube boiler and label its components. (3)
8. 2 kg of an ideal gas occupies a volume of 0.3 m^3 at 10 bar pressure and 500 K temperature. When this gas expands polytropically $pV^{1.2} = C$, the internal energy decreases by 300 kJ. Presuming adiabatic exponent $\gamma = 1.4$, determine: (i) characteristic gas constant, (ii) final temperature and pressure of gas and (iii) heat and work interactions across the system boundary. $[1+(2+2)+(2+1)]$
9. a) 10 kg of wet steam at a pressure of 0.2 MPa is contained in a rigid tank of volume 6.058 m^3 . The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. (3+3)
- b) What do you understand by degree of superheat? What is the value of pressure and temperature at

critical point in case of water?

(1+1)

10. a) Why accessories are essential in boiler? Name two accessories used in boiler and state the function of steam stop valve and fusible plug. ① (1+1+1+1)
- b) A boiler evaporates 5 kg of water per kg of coal burnt on the fire grate in the combustion chamber. The temperature of feed water is 30°C and the steam is raised at 10 bar absolute pressure. Calculate the equivalent evaporation "from and at 100°C " per kg of coal and the efficiency of boiler, if the steam produced is 0.95 dry. Take calorific value of coal as 29800 kJ/kg. (2+2)
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